

Solution 03
Convex Optimization

Exercise 1 **Convexity of Sets**

- 1) The set is $\mathbb{R}^n$, which is convex.
- 2) The set is given by $\{0\}$, which is convex by the definition of a convex set, i.e. $\lambda 0 + (1 - \lambda)0 = 0$
- 3) Same as 2)
- 4) The set is equivalently written as

$$\{x \in \mathbb{R}^n \mid \alpha \leq a^T x\} \cap \{x \in \mathbb{R}^n \mid a^T x \leq \beta\},$$

i.e. as the intersection of two halfspaces. Since a halfspace is a convex set and the intersection preserves convexity we infer that a slab is a convex set.

Exercise 2 **Quadratic Program**

Let us answer both 1) and 2) in one go. The line of reasoning is as follows: We first determine if the optimization problem is convex. If the problem is convex, the KKT conditions would be sufficient for x^* being a minimizer of the problem and this minimizer would also be a global minimizer. (Note that since Slater condition's holds, the KKT conditions are also necessary for x^* being a minimizer.)

In order to analyze convexity of the problem we bring it first to the format used in the lecture, i.e. defining $x := (x_1, x_2, x_3)^T$ we rewrite the objective function as

$$f(x) := \frac{1}{2}x^T Hx + f^T x ,$$

with

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0.1 \end{bmatrix}, \quad f = \begin{bmatrix} 0 \\ 0 \\ 0.55 \end{bmatrix} .$$

Furthermore we obtain the feasible set as

$$X := \{x \mid g(x) := Gx \leq 0, h(x) := a^T x - 1 = 0\} ,$$

where $G = -I_{3 \times 3}$ and $a = \mathbf{1}_{n \times 1}$ is the vector of ones.

Since $f(x)$ is quadratic we can check convexity with the eigenvalues of the Hessian matrix H , which are nonnegative $\{1, 1, 0.1\}$. Also, the functions $g(x)$ and $h(x)$ defining the feasible set are all linear/affine meaning that the feasible set X is convex. Therefore, the optimization problem is convex.

Next, we find that, e.g., $x_0 = (1/3, 1/3, 1/3)^T$ is a strictly feasible point, so, Slater's condition is satisfied.

It remains to verify that point x^* satisfies the KKT conditions. Following the lecture notes the KKT conditions are

$$\begin{aligned} \underbrace{Hx^* + f}_{\nabla f(x^*)} + G^T \lambda^* + a\nu^* &= 0 \\ Gx^* &\leq 0, \quad a^T x^* - 1 = 0 \\ \lambda^* &\geq 0 \\ \lambda_i^* x_i^* &= 0, \quad i \in \{1, 2, 3\}, \end{aligned}$$

which with the data given reads as

$$\begin{bmatrix} 0.5 \\ 0.5 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0.55 \end{bmatrix} - \begin{bmatrix} \lambda_1^* \\ \lambda_2^* \\ \lambda_3^* \end{bmatrix} + \begin{bmatrix} \nu^* \\ \nu^* \\ \nu^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (1)$$

$$\begin{bmatrix} -0.5 \\ -0.5 \\ 0 \end{bmatrix} \leq 0, \quad 0.5 + 0.5 + 0 - 1 = 0 \quad (2)$$

$$\lambda_1^*, \lambda_2^*, \lambda_3^* \geq 0 \quad (3)$$

$$\lambda_1^* 0.5 = 0, \quad \lambda_2^* 0.5 = 0, \quad \lambda_3^* 0 = 0. \quad (4)$$

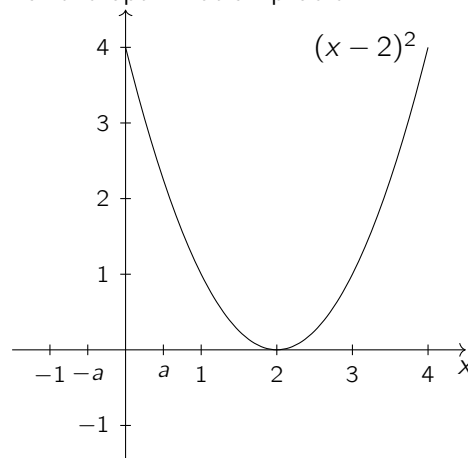
We need to find suitable Lagrange Multipliers $\lambda_1^*, \lambda_2^*, \lambda_3^*$, and ν^* . We conclude from (??) that $\lambda_1^* = \lambda_2^* = 0$. Using these values in (??) gives $\nu^* = -0.5$ and $\lambda_3^* = 0.05$. With these assignments the KKT conditions are satisfied so we conclude that x^* is a global minimizer.

Exercise 3 Non-convex Optimization

- i) The feasible set is given by $x \in \{a, -a\}$. We can evaluate the cost function for all the values of the feasible set. Since $a \in (0, 1) \rightarrow (a-2)^2 \leq (-a-2)^2$. Therefore,

$$x^*(P_1) = a$$

- ii) We can plot the cost function of the optimization problem:



with $a = 0.5$. By inspection, we can find a relaxed optimization problem P_2 :

$$\begin{aligned} P_2 : \min_{x \in \mathbb{R}} & (x-2)^2 \\ \text{subj. to } & |x| \leq a \end{aligned}$$

For the convexified problem P_2 , since Slater's condition holds, the KKT conditions are necessary and sufficient for an optimum point with zero duality gap (See Lec 3, Slide 57). If the KKT conditions lead to $x^*(P_1) = x^*(P_2)$ for all a , then we showed that the solution of the relaxed problem P_2 corresponds to the one of P_1 . For P_2 , the KKT conditions write as follow

- 1) $|x^*| \leq a$
- 2) $\lambda^* \geq 0$
- 3) $(|x^*| - a)\lambda^* = 0$
- 4) $\nabla_x L(x^*, \lambda^*) = 0 = \begin{cases} 2x^* - 4 + \lambda^*, & x^* \geq 0 \\ 2x^* - 4 - \lambda^*, & x^* < 0 \end{cases}$

where the Lagrangian writes $L(x, \lambda) = (x - 2)^2 + \lambda(|x| - a)$. The stationarity (4) condition tell us that the constraint has to be active. Indeed, when $\lambda = 0$ (inactive constraint), the stationarity tell us that $x^* = 2$, which is not primal feasible (1) (contradiction). Therefore, for the KKT conditions to be verified, we have to have $\lambda > 0$.

Then, from the complementary slackness (3), since $\lambda \neq 0$, we have that $(|x^*| - a) = 0$ which gives

$$x^* \in \{-a, a\}.$$

Finally, we can use the stationarity (4) condition for both potential values of x^* . We have that either $\lambda^*(x^* = a) = -2a + 4 > 0$ or $\lambda^*(x^* = -a) = -2a - 4 < 0$. From the dual feasibility (2) condition, it follows that $x^* = a$ is the only primal solution left possible. The solution of P_2 is the same as P_1 .

Exercise 4 Slater's Condition

- i) Slater's condition does not hold. Indeed, none of the feasible solution are strictly feasible.
- ii) The primal solution is given by $x^* = 0$ and $y^* > 0$ by inspection. We have $p^* = 1$. We can write the Lagrangian as

$$L(x, y, \lambda) = e^{-x} + \lambda \frac{x^2}{y}$$

and the dual function is

$$d(\lambda) = \inf_{x, y > 0} e^{-x} + \lambda \frac{x^2}{y} = \begin{cases} 0, & \lambda \geq 0 \\ -\infty, & \lambda < 0 \end{cases}$$

The dual problem writes

$$\begin{aligned} \max \quad & 0 \\ \text{subj. to} \quad & \lambda \geq 0 \end{aligned}$$

and the solution of the dual writes $d^* = 0$ with $\lambda^* \geq 0$. The duality gap is $p^* - d^* = 1$.

- iii) We can verify the KKT conditions with the solution of the primal x^* and y^* and the solution of the dual λ^* .

- 1) $\frac{x^{*2}}{y^*} \leq 0$ and $y^* > 0$
- 2) $\lambda^* \geq 0$
- 3) $\frac{x^{*2}}{y^*} \lambda^* = 0$
- 4) $\begin{bmatrix} -e^{-x^*} \\ 0 \end{bmatrix} + \lambda^* \begin{bmatrix} \frac{2x^*}{y^*} \\ \frac{-x^{*2}}{y^{*2}} \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

The stationarity condition does not hold.